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Robust Multi-UAV Cooperative Trajectory Planning and Power Control for Reliable Communication in the Presence of Uncertain Jammers

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Abstract: Unmanned aerial vehicles (UAVs) have become a promising application for future communication and spectrum awareness due to their favorable features such as low cost, high mobility, and ease of deployment. Nevertheless, the jamming resistance appears to be a new challenge in multi-UAV cooperative communication scenarios. This paper focuses on designing trajectory planning and power allocation for efficient control and reliable communication in a ground control unit (GCU)-controlled UAV network, where the GCU coordinates multi-UAV systems to execute tasks amidst multiple jammers with imperfect location and power information. Specifically, this paper formulates a nonconvex semi-infinite optimization problem to maximize the average worst-case signal-to-interference-plus-noise ratio (SINR) among multiple UAVs by designing robust flight paths and power control strategy under stringent energy and mobility constraints. To efficiently address this issue, this paper proposes a powerful iterative algorithm utilizing the S-procedure and the successive convex approximation (SCA) method. Extensive simulations validate the effectiveness of the proposed strategy.

Keywords: multi-UAV; nonconvex optimization; trajectory planning; power allocation

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1. Introduction

The future information-driven battlefields will be characterized by complex electromagnetic conditions, extensive mobility, and vast data capacities [1,2]. However, traditional electronic spectrum technologies are no longer sufficient to meet the demands of modern warfare. As a low-altitude information platform, unmanned aerial vehicles (UAVs) can complement and improve existing communication networks through their exceptional mobility, easy deployment, and various other benefits [3–6]. First of all, the communication quality is significantly improved due to the predominantly line-of-sight (LoS) wireless channel from the ground control unit (GCU) to the UAVs which generally results in low path loss. Additionally, multiple UAVs operating as an intelligent swarm can greatly enhance operational effectiveness through coordinated manners. Due to these desirable features, the applications of multi-UAV cooperative communication and spectrum awareness play a crucial role in future battlefields [7,8].

However, UAV swarms frequently encounter external interference from sources such as radiation stations and enemy radar [9] which complicates the coordinated planning and degrades the communication performance. Motivated by these challenges, numerous studies have focused on exploring solid technology for multi-UAV communication

systems [2,10,11]. Specifically, several efforts aimed at optimizing UAV swarms' communication performance have primarily concentrated on two key areas: improving the communication link quality and bolstering resistance to interference from external jammers. In particular, the authors of [12] conducted a systematic study on the joint optimization problem of multiple UAV trajectories. This research aimed to maximize the average throughput of the UAV relay system while considering constraints on communication resources and swarm maneuverability, and subsequently proposed an effective algorithm to optimize this problem. Additionally, in [13], a collaborative control method was investigated to aid search and rescue operations in disaster scenarios. This method achieved the optimization of swarm flight trajectories and route-based relay communication under different modes of UAV swarm search and relay states, significantly enhancing the system's communication performance. To ensure reliable signal transmission in multi-UAV communication systems, the authors of [14] developed an integrated optimization approach that includes channel assignment, power allocation, and trajectory planning for UAV swarms. Moreover, an efficient resource allocation strategy for multi-UAV cooperative communication in the presence of multiple interference styles is proposed in [15], where the authors designed a priority-aware resource coordination method to mitigate interferences and enhance network communication quality.

The studies mentioned above primarily focus on enhancing the system's data transmission capacity through the joint optimization of UAV trajectories and resource allocation, while overlooking the impact of external interference. To explore this issue, the authors in [16,17] conducted research on interference resistance technologies in multi-UAV systems. Specifically, the authors in [16] considered a scenario where rotary-wing UAVs assist multiple ground users in communication. This approach employed frequency division multiplexing and trajectory optimization to reduce the interference signals and improve service quality. In [17], the authors expanded on the communication scenario in [16] by considering the joint optimization of UAV trajectories, speed, acceleration, and altitude. This approach aimed to reduce interference evoked by the radiation stations, thereby further enhancing the system's communication performance. Furthermore, some studies [18,19] have studied the joint optimization of UAV swarm trajectories and communication in scenarios with uncertain external interference, an approach which is more suitable for realistic scenarios. In [18], the authors proposed an efficient method for joint optimization of UAV swarms' trajectories and power under uncertain eavesdropper location information to maximize the secrecy rate in UAV-assisted cognitive radio networks. Meanwhile, the authors in [19] explored the maximization of throughput in UAV-assisted communication systems, where multiple users communicate with UAV swarms in scenarios where eavesdroppers' locations are uncertain. It ensures the required quality of service (QoS) for each user by optimizing the UAVs' trajectories and transmission power. To meet diverse QoS requirements across different applications, the authors in [20] investigated the problem of maximizing throughput in a UAV-assisted communication system. They analyzed a scenario where multiple ground users interact with a multi-UAV swarm while contending with jammers that have uncertain location information.

Although the aforementioned works have contributed to the trajectory planning and resource allocation design for multi-UAV communication systems, several issues remain unaddressed. First, existing studies primarily address the uncertainty in the locations of jammers or eavesdroppers. In practice, jammers often use tactics like electromagnetic silence or shielding, making it challenging to estimate their transmission power. Additionally, existing literature rarely explores the energy constraints of a UAV swarm. In fact, energy consumption significantly impacts the efficiency of mission execution [21]. Motivated by the aforementioned concerns, this paper focuses on trajectory planning and power allocation for efficient control and reliable communication within a GCU-controlled UAV network. We examine a realistic scenario where the GCU directs multiple UAVs to execute tasks amidst numerous jammers. The UAVs have partial knowledge of the jammers' power and locations, making the quality of the wireless link heavily dependent

on the UAVs' trajectories over time. The goal of this study is to develop robust flight paths and power control strategies to counteract the jammers' effects and enhance the quality of control signals. Nonetheless, the optimization problem is highly challenging due to its nonconvex and semi-infinite nature. The main contributions are summarized as follows:

In the scenario where the power and location information of jammers are uncertain, we present a novel multi-UAV trajectory and communication optimization model. Our objective is to maximize the average worst-case SINR received by the UAV swarm through joint optimization of UAV flight parameters and transmission power. This problem poses a nonconvex, semi-infinite optimization challenge with nonlinear couplings among variables, making it difficult to obtain the optimal solution.

To handle the intractable problem, we introduce slack variables and employ the S-procedure algorithm to transform the original problem into two manageable convex subproblems. By leveraging the block coordinate descent (BCD) and successive convex approximation (SCA) algorithms, we alternately optimize these subproblems and, ultimately, obtain a powerful solution to the formulated problem.

We conduct comprehensive experiments across various parameter configurations. The simulation results clearly indicate that our proposed algorithm can effectively optimize the UAV flight paths and control power. Compared to other benchmark algorithms, our algorithm notably enhances the communication performance of the UAV swarms.

The rest of this paper is structured as follows. Section 2 introduces the robust trajectory planning and power control problems. Section 3 presents a low-complexity iterative algorithm to address this problem. Simulation results are discussed in Section 4. Finally, conclusions are drawn in Section 5.

2. System Method

Consider an uplink multi-UAV wireless communication system as depicted in Figure 1, where a GCU controls $M \geq 1$ UAVs to perform specific tasks with $J \geq 1$ jammers deployed throughout the area. During this mission, the GCU regularly sends control signals to the UAVs. Meanwhile, multiple jammers transmit interference signals to disrupt these control links.

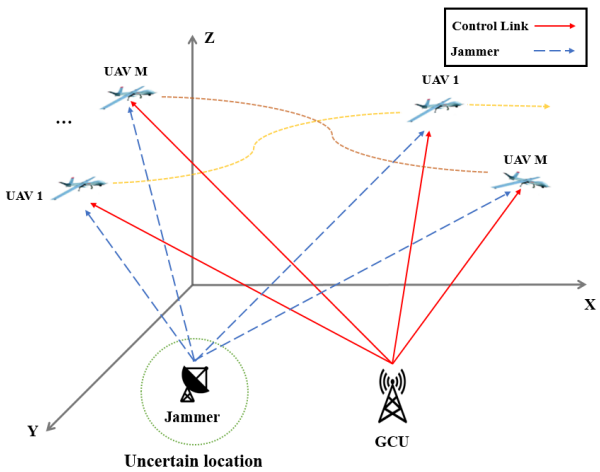

Figure 1. Illustration of an uplink multi-UAV communication scenario where a GCU manages multiple UAVs amidst several jammers with imperfect information.

This considers a three-dimensional (3D) coordinate system, assuming the location of the GCU is $\left[\mathbf{w}_s^T, 0\right]^T \in \mathbb{R}^{3\times 1}$ and the location of the jammers ($j \in \{1, 2, 3, \ldots, J\}$) is $\left[\mathbf{w}_j^T, 0\right]^T \in \mathbb{R}^{3\times 1}$, where $\mathbf{w}_s \in [x_s, y_s]^T$ and $\mathbf{w}_j \in \left[x_j, y_j\right]^T$ represent the horizontal position of the GCU and jammers, respectively. The aim of this paper is to optimize the flight trajectories of the UAV swarm within the time duration T, as well as the control power exerted by the GCU over the UAV swarm, to reduce interference from jammers on the

communication links and enhance communication quality between the GCU and the UAV swarm. Before the problem formulation, we introduce models for uncertain information about jammers, UAV trajectories, GCU-to-UAV transmission channels, as well as the energy consumption.

2.1. Uncertain Jammer Model

The optimization of UAV trajectories relies on accurately estimating information from both the GCU and potential jammers. According to research in [18,19], UAV can easily acquire the precise state of the GCU through information exchange. However, accurately obtaining the information about jammers poses challenges due to potential errors. UAV can estimate the approximate locations of jammers by using cameras or synthetic aperture radar (SAR) [20] and estimate their approximate power with the aid of the carrier detection technology. Thus, the bounded error model can be employed to describe the relationship between the actual and estimated information of the jammers. Specifically, the location of the jammers can be modeled as:

$$\begin{gathered} \mathbf{w}_j = \tilde{\mathbf{w}}_j + \Delta\mathbf{w}_j, \forall j \\ \mathcal{A}_j \triangleq \left\{ \Delta\mathbf{w}_j = \left[\Delta x_j, \Delta y_j\right]^T \in \mathbb{R}^{2\times 1} : \|\Delta\mathbf{w}_j\|^2 \leq \epsilon_j \right\}, \end{gathered} \tag{1}$$

where $\tilde{\mathbf{w}}_j$ and $\Delta\mathbf{w}_j$ denote the approximate location and location error for the j-th jammer, respectively, and ϵ_j denotes the bound of the location estimation error. Similarly, the power of the jammers can be represented as:

$$\begin{gathered} p_j = \widetilde{p}_j + \Delta p_j, \forall j \\ \psi_j \triangleq \left\{ \Delta p_j \in \mathbb{R} : \|\Delta p_j\| \leq \xi_j \right\}, \end{gathered} \tag{2}$$

where $\widetilde{p}_j$ and Δp_j represent the approximate power and power error for the j-th jammer, respectively, and ξ_j expresses the bound of the power estimation error.

2.2. UAV Flight Model

To adapt to complex electromagnetic environments, the UAV swarm must have dynamic adjustment capabilities. This subsection establishes the flight model for the UAV swarm. Assuming all UAVs ($m \in \{1, 2, 3, \ldots, M\}$) fly at a fixed appropriate altitude H, which starts from the certain initial positions $\left(\mathbf{q}_{m,I} = [x_{m,I}, y_{m,I}]^T \in \mathbb{R}^{2\times 1}\right)$ to the destinations $\left(\mathbf{q}_{m,F} = [x_{m,F}, y_{m,F}]^T \in \mathbb{R}^{2\times 1}\right)$. For the sake of analysis, we discretized the total flight time T into N time slots. As N approaches infinity, the duration of each time slot $\delta = T/N$ becomes small enough that the positions of the multiple UAVs can be considered nearly constant within each slot. Therefore, at the N-th moment, the horizontal trajectory of the m-th UAV can be expressed as $\mathbf{q}_m\ [n] = [x_m[n], y_m[n]]^T \in \mathbb{R}^{2\times 1}$. Moreover, we let $\mathbf{u}_m\ [n] = \left[\mathrm{u}_{m,x}[n], \mathrm{u}_{m,y}[n]\right]^T \in \mathbb{R}^{2\times 1}$ and $\mathbf{a}_m\ [n] = \left[\mathrm{a}_{m,x}[n], \mathrm{a}_{m,y}[n]\right]^T \in \mathbb{R}^{2\times 1}$ denote the speed vector and acceleration vector of the m-th UAV, respectively. According to the principles of physics, the UAVs' flight model can be described as follows:

$$\begin{gathered} \mathbf{q}_m\ [n] = \mathbf{q}_m\ [n-1] + \mathbf{u}_m\ [n-1]\delta + \tfrac{1}{2}\mathbf{a}_m\ [n-1]\delta^2, \quad n = 2, 3, \ldots, N, \forall m, \\ \mathbf{u}_m\ [n] = \mathbf{u}_m\ [n-1] + \mathbf{a}_m\ [n-1]\delta, \quad n = 2, 3, \ldots, N, \forall m, \\ \mathbf{q}_m\ [1] = \mathbf{q}_{m,I} + \mathbf{u}_{m,I}\delta + \tfrac{1}{2}\mathbf{a}_{m,I}\delta^2, \\ \mathbf{q}_m\ [N] = \mathbf{q}_{m,\mathrm{F}}, \mathbf{u}_m\ [N] = \mathbf{u}_{m,\mathrm{F}}, \forall m, \\ \|\mathbf{a}_m\ [n]\ \| \leq a_{\max}, \forall n, m, \\ \|\mathbf{u}_m\ [n]\ \| \leq u_{\max}, \ \forall n, m, \\ \|\mathbf{u}_m\ [n]\ \| \geq u_{\min}, \ \forall n, m, \end{gathered} \tag{3}$$

where $\mathbf{a}_{m,I}$, $\mathbf{u}_{m,I}$, and $\mathbf{q}_{m,I}$ denote the initial acceleration, speed, and location, respectively, $\mathbf{u}_{m,\mathrm{F}}$ and $\mathbf{q}_{m,\mathrm{F}}$ represent the final speed and location, respectively, and $a_{\max}$, $u_{\max}$, and $u_{\min}$ express the maximum acceleration, maximum speed, and minimum speed, respectively.

2.3. Channel Gain Model

The trajectory and power optimization of the UAV swarm depend on advanced knowledge of channel state information (CSI). Existing research [22–25] indicates that the communication channels between the GCU and the UAVs is dominated by LoS links. Consequently, this paper utilizes the free space path loss model to characterize the communication channel gain between UAVs and GCU. Specifically, the channel gain between the m-th UAV and the GCU in the n-th time slot is represented as:

$$\mathbf{h}_{m,s}[n] = \beta_0 d_{m,s}^{-\alpha}[n] = \frac{\beta_0}{(||\mathbf{q}_m\ [n] - \mathbf{w}_s||^2 + H^2)^{\alpha/2}}, \forall m, \tag{4}$$

where $d_{m,s} = \sqrt{||\mathbf{q}_m\ [n] - \mathbf{w}_s||^2 + H^2}$ represents the distance from the m-th UAV to the GCU and β_0 expresses the channel gain in a reference distance ($d_0 = 1m$) [12]. The term α represents the path loss factor, where, typically, $\alpha \geq 2$ [19,20,26–31].

In a similar way, the channel gain between the j-th jammer and the m-th UAV in the n-th time slot can be expressed as:

$$\mathbf{h}_{mj}[n] = \beta_0 d_{mj}^{-\alpha}[n] = \frac{\beta_0}{(||\mathbf{q}_m\ [n] - \mathbf{w}_j||^2 + H^2)^{\alpha/2}}, \forall m, j, \tag{5}$$

where $d_{mj} = \sqrt{||\mathbf{q}_m\ [n] - \mathbf{w}_j||^2 + H^2}$ denotes the distance between the j-th jammer and the m-th UAV in the n-th time slot.

2.4. Energy Consumption Model

During the task execution phase, the energy consumption of the UAV was mainly divided into two components. One part is the communication energy, which is required to maintain signal interaction between the UAV and the GCU/other UAVs. The other part is flight energy, which is necessary to sustain the flight of the UAV at a high altitude. According to [12], the communication energy consumption of a UAV is much lower than its flight energy consumption and is typically negligible. Therefore, the energy consumption of the m-th UAV over N time slots can be formulated as:

$$E_m = \delta \sum_{n=1}^{N} \left[c_1 ||\mathbf{u}[n]||^3 + \frac{c_2}{||\mathbf{u}[n]||} \left(1 + \frac{||\mathbf{a}[n]||^2}{g^2} \right) \right] + \Delta_k, \forall m \tag{6}$$

Here, c_1 and c_2 represent aerodynamic parameters, while g denotes the gravitational acceleration. The term $\Delta_k = 0.5 g_m \left(\mathbf{u}_{m,\mathrm{F}}^2 - \mathbf{u}_{m,I}^2 \right)$ indicates the change in kinetic energy of the m-th UAV, where g_m denotes the mass of the m-th UAV.

2.5. Power-Constrained Model

The power management of the GCU over the UAV swarm $p_m\ [n]$ is restricted by total power constraints and individual transmission power limits [31]. Hence, this constraint can be formulated as:

$$\begin{aligned} \sum_{m=1}^{M} p_m\ [n] \leq p_{\max}, \forall n \\ p_m\ [n] \leq p_0, \forall m, n \end{aligned} \tag{7}$$

where $p_{\max}$ represents the maximum transmission power of the GCU across all UAVs within a single time slot and p_0 denotes the maximum transmission power of the GCU for any individual UAV in an individual time slot.

2.6. Problem Formulation

In this subsection, we present the problem formulation of the multi-UAV trajectory optimization and power control. Given the incomplete information on each jammer, the worst-case received SINR of the UAV at a specific time slot can be expressed as:

$$\gamma_m[n] = \min_{\substack{\Delta \mathbf{w}_j \in \mathcal{A}_j, \\ \Delta p_j \in \psi_j}} \frac{p_m[n] h_{m,s}[n]}{\sum_{j=1}^{J} p_j h_{m,j}[n] + \sigma^2}, \tag{8}$$

where σ^2 denotes the power of the additive white Gaussian noise (AWGN) at the receiver and p_j represents the transmit power of the j-th jammer.

To simplify the explanation, we defined $\mathbf{Q} \triangleq \{\mathbf{q}_m[n], \forall m, n\}$, $\mathbf{U} \triangleq \{\mathbf{u}_m[n], \forall m, n\}$, and $\mathbf{A} \triangleq \{\mathbf{a}_m[n], \forall m, n\}$ as the sets representing the trajectories, speeds, and accelerations of the UAV swarm, respectively. We also denoted $\mathbf{P} \triangleq \{p_m[n], \forall m, n\}$ as the power allocation of the GCU. Our objective was to jointly optimize the trajectories $\mathbf{Q}$, speeds $\mathbf{U}$, and accelerations $\mathbf{A}$ of the UAV swarm, as well as the transmit power $\mathbf{P}$ of the GCU, to maximize the average worst-case SINR of the receivers while they are subject to the UAV swarm's flight and energy constraints, as well as the power limitations of the GCU. The problem can be formulated as follows:

$$(\text{P1}) \max_{\{\mathbf{Q},\mathbf{U},\mathbf{A},\mathbf{P}\}} \frac{1}{N} \frac{1}{M} \sum_{n=1}^{N} \sum_{m=1}^{M} \min_{\substack{\Delta \mathbf{w}_j \in \mathcal{A}_j, \\ \Delta p_j \in \psi_j}} \frac{p_m[n] h_{m,s}[n]}{\sum_{j=1}^{J} p_j h_{mj}[n] + \sigma^2}, \tag{9a}$$

$$s.t.\ \delta \sum_{n=1}^{N} \left[c_1 ||\mathbf{u}_m[n]||^3 + \frac{c_2}{||\mathbf{u}_m[n]||} \left(1 + \frac{||\mathbf{a}_m[n]||^2}{g^2} \right) \right] + \Delta_k \leq \Gamma, \forall m, \tag{9b}$$

$$\mathbf{q}_m[n] = \mathbf{q}_m[n-1] + \mathbf{u}_m[n-1]\delta + \frac{1}{2}\mathbf{a}_m[n-1]\delta^2,\ n = 2, 3, \ldots, N, \forall m, \tag{9c}$$

$$\mathbf{u}_m[n] = \mathbf{u}_m[n-1] + \mathbf{a}_m[n-1]\delta,\ n = 2, 3, \ldots, N, \forall m, \tag{9d}$$

$$\mathbf{q}_m[1] = \mathbf{q}_{m,I} + \mathbf{u}_{m,I}\delta + \frac{1}{2}\mathbf{a}_{m,I}\delta^2, \forall m, \tag{9e}$$

$$\mathbf{q}_m[N] = \mathbf{q}_{m,F}, \mathbf{u}_m[N] = \mathbf{u}_F, \forall m \tag{9f}$$

$$\|\mathbf{a}_m[n]\| \leq a_{\max}, \forall n, m\ , \tag{9g}$$

$$\|\mathbf{u}_m[n]\| \leq u_{\max}, \forall n,\ m, \tag{9h}$$

$$\|\mathbf{u}_m[n]\| \geq u_{\min}, \forall n,\ m, \tag{9i}$$

$$\|\mathbf{q}_m[n] - \mathbf{q}_k[n]\| \leq d_{\min}^2, \forall n, m \neq k\ , \tag{9j}$$

$$\sum_{n=1}^{N} p_m[n] \leq p_{\max}, \forall n, \tag{9k}$$

$$p_m[n] \leq p_0\ , \forall n, m, \tag{9l}$$

where (9b) denotes the energy constraints of the UAVs (assuming all UAVs have a limit energy consumption Γ), (9c)–(9f) represent their mobility constraints, and (9g)–(9i) indicate the UAVs' maximum speed, minimum speed, and maximum acceleration, respectively. The equation (9j) expresses the collision constraints between any two UAVs, while (9k) and (9l) refer to the total power and the single transmission power constraints of the GCU, respectively.

Based on the above model, it was found that the variables in problem (P1) were highly coupled. The optimal solution posed significant challenges due to two main reasons. Firstly, the nonlinear objective function and the constraints in (9b) and (9i) made the formulation both nonconvex and nondifferentiable. Additionally, the semi-infinite inequality constraint introduced by the uncertainty in the jammers' information further complicated the problem. In the following section, we transform the formulation in several ways and propose an efficient algorithm to address it.

3. Proposed Solution

In this section, we employ the BCD algorithm to divide the original problem into two subproblems for iterative solving. Specifically, this involves optimizing the GCU's transmission power $\mathbf{P}$ given the UAV flight parameters $\{\mathbf{Q}, \mathbf{U}, \mathbf{A}\}$, and then optimizing the UAV flight parameters $\{\mathbf{Q}, \mathbf{U}, \mathbf{A}\}$ given a certain transmission power $\mathbf{P}$ of the GCU. This process is repeated iteratively until the algorithm converges. The detailed subproblems are presented below.

3.1. Power Optimization

When the UAVs' flight parameters $\{\mathbf{Q}, \mathbf{U}, \mathbf{A}\}$ are determined, the power allocation problem can be expressed as follows:

$$\text{(P2)} \max_{\{\mathbf{P}\}} \frac{1}{N}\frac{1}{M}\sum_{n=1}^{N}\sum_{m=1}^{M} \min_{\substack{\Delta \mathbf{w}_j \in \mathcal{A}_j, \\ \Delta p_j \in \psi_j}} \frac{p_m[n]h_{m,s}[n]}{\sum_{j=1}^{J} p_j h_{mj}[n] + \sigma^2} \tag{10a}$$

$$s.t. \sum_{n=1}^{M} p_m[n] \leq p_{\max}, \forall n, \tag{10b}$$

$$p_m[n] \leq p_0 \ , \forall n, m, \tag{10c}$$

To simplify the problem (P2), we introduced an auxiliary variable η, resulting in the following equivalent formulation:

$$\text{(P3)} \max_{\{\mathbf{P},\eta\}} \eta \tag{11a}$$

$$s.t. \frac{1}{N}\frac{1}{M}\sum_{n=1}^{N}\sum_{m=1}^{M} \min_{\substack{\Delta \mathbf{w}_j \in \mathcal{A}_j, \\ \Delta p_j \in \psi_j}} \frac{p_m[n]h_{m,s}[n]}{\sum_{j=1}^{J} p_j h_{mj}[n] + \sigma^2} \geq \eta \tag{11b}$$

The constraint (11b) involves uncertain jammer parameters. The analysis indicates that, given the UAVs' flight parameters, the jammers select the maximum transmission power (i.e., $\widetilde{p}_j + \xi_j$) and the minimum distance to the UAV (i.e., $\sqrt{\left|\left|\mathbf{q}_m \ [n] - \tilde{\mathbf{w}}_j - \epsilon_j\right|\right|^2}$) for each time slot. This approach eliminates the 'min' term from the constraint (11b), thereby simplifying the problem (P3).

Through the above transformation, the problem (P3) can be regarded as a standard convex optimization problem, which can be efficiently solved using convex optimization tools such as CVX.

3.2. Flight Parameters Optimization

Given the GCU's transmission power $\mathbf{P}$, the optimization problem for UAV swarm flight parameters can be defined as follows:

$$\text{(P4)} \max_{\{\mathbf{Q},\mathbf{U},\mathbf{A}\}} \frac{1}{N}\frac{1}{M}\sum_{n=1}^{N}\sum_{m=1}^{M} \min_{\substack{\Delta \mathbf{w}_j \in \mathcal{A}_j, \\ \Delta p_j \in \psi_j}} \frac{p_m[n]h_{m,s}[n]}{\sum_{j=1}^{J} p_j h_{mj}[n] + \sigma^2}, \tag{12a}$$

$$s.t.\ \delta\sum_{n=1}^{N}\left[c_1||\mathbf{u}_m[n]||^3+\frac{c_2}{||\mathbf{u}_m[n]||}\left(1+\frac{||\mathbf{a}_m[n]||^2}{g^2}\right)\right]+\Delta_k\leq\Gamma,\forall m, \tag{12b}$$

$$\mathbf{q}_m[n]=\mathbf{q}_m[n-1]+\mathbf{u}_m[n-1]\delta+\frac{1}{2}\mathbf{a}_m[n-1]\delta^2,\ n=2,3,\ldots,N,\ \forall m, \tag{12c}$$

$$\mathbf{u}_m[n]=\mathbf{u}_m[n-1]+\mathbf{a}_m[n-1]\delta,\ n=2,3,\ldots,N,\forall m, \tag{12d}$$

$$\mathbf{q}_m[1]=\mathbf{q}_{m,I}+\mathbf{u}_{m,I}\delta+\frac{1}{2}\mathbf{a}_{m,I}\delta^2,\forall m, \tag{12e}$$

$$\mathbf{q}_m[N]=\mathbf{q}_{m,F},\mathbf{u}_m[N]=\mathbf{u}_F,\forall m \tag{12f}$$

$$\|\mathbf{a}_m[n]\ \|\leq a_{\max},\forall n,m\ , \tag{12g}$$

$$\|\mathbf{u}_m[n]\ \|\leq u_{\max},\ \forall n,\ m, \tag{12h}$$

$$\|\mathbf{u}_m[n]\ \|\geq u_{\min},\ \forall n,\ m, \tag{12i}$$

$$\|\mathbf{q}_m[n]-\mathbf{q}_k[n]\ \|\leq d_{\min^2},\ \forall n,m\neq k, \tag{12j}$$

Due to the complexity of the objective function and the difficulty associated with the handling of semi-infinite and nonconvex constraints, the problem (P4) is challenging to optimize directly. In the following section, we introduce a powerful algorithm to address it. Initially, we simplify the nonlinear objective function into a more manageable form. Subsequently, to handle the semi-infinite constraints arising from the uncertain information on the jammers, we offer insights to transform these complex constraints into an equivalent and more explicit form using the S-procedure algorithm. To tackle the nonconvex problem, we develop an iterative solution approach that employs slack variables and the SCA method. Finally, we provide a summary of the algorithm.

3.2.1. Reformulation of the Objective Function in (P3)

As seen in (12a), the higher the power of the jammers, the worse the SINR received by the UAVs. Therefore, to simplify the problem, we can substitute p_j with $(\widetilde{p}_j+\xi_j)$, thereby removing the variable Δp_j from the objective function. Additionally, since the location uncertainty variable Δw_j for each radiation source only appears in the denominator of the objective function, inputting (4) and (5) into the objective function allows problem (P1) to be equivalently transformed as follows:

$$\text{(P5)}\ \max_{\{\mathbf{Q},\mathbf{U},\mathbf{A}\}}\frac{1}{N}\frac{1}{M}\sum_{n=1}^{N}\sum_{m=1}^{M}\frac{p_m[n]\dfrac{\beta_0}{(||\mathbf{q}_m\ \ [n]-\mathbf{w}_s||^2+H^2)^{\alpha/2}}}{\sum\limits_{j=1}^{J}(\widetilde{p}_j+\xi_j)\dfrac{\beta_0}{\min\limits_{\Delta\mathbf{w}_j\in\mathcal{A}_j}(||\mathbf{q}_m\ \ [n]-\mathbf{w}_j||^2+H^2)^{\alpha/2}}+\sigma^2}, \tag{13a}$$

$$s.t.\ (12b)-(12j) \tag{13b}$$

The challenge of problem (P5) stems from the complexity of its objective function. To simplify the problem, we introduced the following lemma:

Lemma 1. *By introducing the auxiliary variables* $\boldsymbol{I}\triangleq[I_m[n],\forall m,n]$ *and* $\boldsymbol{L}\triangleq[L_m[n],\forall m,n]$, *problem (P5) can be equivalently reformulated as follows:*

$$\text{(P6)}\ \max_{\{\mathbf{Q},\mathbf{U},\mathbf{A},\mathbf{I},\mathbf{L}\}}\frac{1}{N}\frac{1}{M}\sum_{n=1}^{N}\sum_{m=1}^{M}\frac{p_m[n]\beta_0}{\mathrm{I}_m[n]\mathrm{L}_m[n]} \tag{14a}$$

$$s.t.\,(||\mathbf{q}_m\ \ [n]-\mathbf{w}_s||^2+H^2)^{\alpha/2}\leq\mathrm{I}_m[n],\forall m,n, \tag{14b}$$

$$\sum_{j=1}^{J}\left(\widetilde{p}_j+\xi_j\right)\frac{\beta_0}{\min\limits_{\Delta\mathbf{w}_j\in\mathcal{A}_j}\left(||\mathbf{q}_m\ [n]-\mathbf{w}_j||^2+H^2\right)^{\alpha/2}}+\sigma^2\leq L_m[n],\forall m,n, \tag{14c}$$

$$(12b)-(12j) \tag{14d}$$

Proof. This lemma can be proven by the following contradiction:

Assuming that both the equations $(||\mathbf{q}_m\ [n]-\mathbf{w}_s||^2+H^2)^{\alpha/2}=\mathrm{I}_m[n]$, $\forall m,n$ and $\sum\limits_{j=1}^{J}\left(\widetilde{p}_j+\xi_j\right)\frac{\beta_0}{\min\limits_{\Delta\mathbf{w}_j\in\mathcal{A}_j}\left(||\mathbf{q}_m\ [n]-\mathbf{w}_j||^2+H^2\right)^{\alpha/2}}+\sigma^2=L_m[n],\forall m,n$ hold simultaneously, the optimal solutions to the optimization problems (P5) and (P6) are equivalent. However, if there exists an optimal solution that satisfies the constrains $(||\mathbf{q}_m\ [n]-\mathbf{w}_s||^2+H^2)^{\alpha/2}<\mathrm{I}_m[n],\forall m,n$ or $\sum\limits_{j=1}^{J}\left(\widetilde{p}_j+\xi_j\right)\frac{\beta_0}{\min\limits_{\Delta\mathbf{w}_j\in\mathcal{A}_j}\left(||\mathbf{q}_m\ [n]-\mathbf{w}_j||^2+H^2\right)^{\alpha/2}}$ $+\sigma^2<L_m[n],\forall m,n$, we can reduce the variables $I_m[n]$ and $L_m[n]$ to improve the objective function value in (P6). This contradicts the optimal solution in (P6). Therefore, the optimal solutions to problems (P5) and (P6) are identical. This lemma is proven.

Furthermore, the term $(I_m[n]L_m[n])^{-1}$ in the objective function of problem (P6) renders the problem nonconvex. To simplify the objective function, we can use the following lemma. □

Lemma 2. *For any feasible point $(I_{m,\text{fea}}[n]L_{m,\text{fea}}[n])$, the lower bound of the terms$(I_m[n]L_m[n])^{-1}$can be expressed as:*

$$\begin{aligned}\widetilde{\gamma}_{m,\ \text{fea}}\ [n]=&\frac{1}{I_{m,\ \text{fea}}\ [n]L_{m,\ \text{fea}}\ [n]}-\frac{1}{I^2_{m,\ \text{tat}}\ [n]L_{m,\ \text{fea}}\ [n]}(I_m[n]-I_{m,\ \text{fea}}\ [n])\\&-\frac{1}{L^2_{m,\ \text{fes}}\ [n]I_{m,\ \text{fea}}\ [n]}(L_m[n]-L_{m,\ \text{fea}}\ [n]).\end{aligned} \tag{15}$$

Proof. Since the function $f(x,y)=\frac{1}{xy},(x,y>0)$ is jointly convex in x and y, the first-order Taylor expansion of a convex function provides a global lower bound. Therefore, at a given feasible point $(x_{\text{fea}},y_{\text{fea}})$, the following relationship holds:

$$\frac{1}{xy}\geq\frac{1}{x_{\text{fea}}y_{\text{fea}}}-\frac{1}{x^2_{\text{fea}}y_{\text{fea}}}(x-x_{\text{fea}})-\frac{1}{y^2_{\text{fea}}x_{\text{fea}}}(y-y_{\text{fea}}). \tag{16}$$

By setting $x=I_{m,\text{fea}}[n]$ and $y=L_{m,\text{fea}}[n]$, the Lemma 2 can be proven. □

3.2.2. Reformulation of the Semi-Infinite Constraint in (14c)

Owing to the uncertainty in the positions of the jammers, constraint (14c) is difficult to address directly. By introducing the auxiliary variable $\mathbf{S}=\left\{s_{mj}[n],\forall n,m,j\right\}$, constraint (14c) can be equivalently transformed into the following two constraints:

$$\sum_{j=1}^{J}\left(\widetilde{p}_j+\xi_j\right)\frac{\beta_0}{\left(s_{mj}[n]\right)^{\alpha/2}}+\sigma^2\leq L[n],\forall m,n, \tag{17a}$$

$$\min_{\Delta\mathbf{w}_j\in\mathcal{A}_j}\left|\left|\mathbf{q}_m\ [n]-\mathbf{w}_j\right|\right|^2+H^2\geq s_{mj}[n],\forall m,n,j, \tag{17b}$$

Clearly, constraint (17a) is convex. However, constraint (17b) involves infinite $\left(\Delta x_j,\Delta y_j\right)$ pairs, making it difficult to tackle. To simplify the problem, we can input (1) into the semi-infinite constraint (17a), so as to obtain the following constraints:

$$\Delta x_j^2+\Delta y_j^2-\epsilon_j^2\leq 0,\forall j, \tag{18a}$$

$$-\left(x_m[n]-\Delta x_j-\widetilde{x}_j\right)^2-\left(y_m[n]-\Delta y_j-\widetilde{y}_j\right)-H_j+s_{mj}[n]\leq 0,\forall n,m,j. \tag{18b}$$

To address constraints (18a) and (18b), we can use the following lemma:

Lemma 3. *(S-procedure) [32]: for the function* $f_i(\mathbf{x})=\mathbf{x}^T\mathbf{A}_i\mathbf{x}+\mathbf{b}_i\mathbf{x}+c_i, i\in\{1,2\}$, *where* $\mathbf{x}\in\mathbb{R}^{M\times 1}, \mathbf{A}_i\in\mathbb{H}^M, \mathbf{b}_i\in\mathbb{R}^{M\times 1}$, *and* $c_i\in\mathbb{R}$, *the condition* $f_1(\mathbf{x})\leq 0\Rightarrow f_2(\mathbf{x})\leq 0$ *holds if, and only if, there exists* $\lambda\geq 0$ *such that:*

$$\lambda\begin{bmatrix}\mathbf{A}_1 & \mathbf{b}_1\\ \mathbf{b}_1^H & c_1\end{bmatrix}-\begin{bmatrix}\mathbf{A}_2 & \mathbf{b}_2\\ \mathbf{b}_2^H & c_2\end{bmatrix}\geq 0 \tag{19}$$

In accordance with Lemma 3, the condition $(18a)\Rightarrow(18b)$ holds if, and only if, there exists $\lambda_{m,j}[n]$ that satisfies the following:

$$\left(\mathbf{\Phi}x_m[n],y_m[n],s_{mj}[n],\lambda_{mj}[n]\right),\forall n,m,j \tag{20}$$

where:

$$\mathbf{\Phi}(x_m[n],y_m[n],s_{mj}[n],\lambda_{mj}[n])=\begin{bmatrix}\lambda_{mj}[n]+1 & 0 & \widetilde{x}_j-x_m[n]\\ 0 & \lambda_{mj}[n]+1 & \widetilde{y}_j-y_m[n]\\ \widetilde{x}_j-x_m[n] & \widetilde{y}_j-y_m[n] & -\varepsilon_j^2\lambda_{mj}[n]+c_{mj}[n]\end{bmatrix}$$ and

$c_{mj}[n]=x_m^2[n]-2x_m[n]\widetilde{x}_j+\widetilde{x}_j^2+y_m^2[n]-2y_m[n]\widetilde{y}_j+\widetilde{y}_j^2+H^2-s_{mj}[n]$.

By introducing the auxiliary variable $\boldsymbol{\lambda}=\left\{\lambda_{mj}[n],\forall n,m,j\right\}$, the semi-infinite constraint (17b) can be equivalently represented as the following two constraints:

$$\mathbf{\Phi}=\left(x_m[n],y_m[n],s_{m,j}[n],\lambda_{m.j}[n]\right)\geq 0,\forall n,m,j, \tag{21a}$$

$$\lambda_{m.j}[n]\geq 0,\forall n,m,j. \tag{21b}$$

3.2.3. Reformulation of Nonconvex Constraints

By scaling the objective function to the original function's lower bound and equivalently transforming the semi-infinite constraints, problem (P6) can be reformulated as:

$$\text{(P7)}\quad \max_{\{\mathbf{Q},\mathbf{U},\mathbf{A},\mathbf{I.L},\mathbf{S},\boldsymbol{\lambda}\}}\frac{1}{N}\frac{1}{M}\sum_{n=1}^{N}\sum_{m=1}^{M}p_m[n]\beta_0\widetilde{\gamma}_{m,\text{fea}}[n] \tag{22a}$$

$$s.t.\delta\sum_{n=1}^{N}\left[c_1||\mathbf{u}_m[n]||^3+\frac{c_2}{||\mathbf{u}_m[n]||}\left(1+\frac{||\mathbf{a}_m[n]||^2}{g^2}\right)\right]+\Delta_k\leq\Gamma,\forall m, \tag{22b}$$

$$\|\mathbf{u}_m[n]\|\geq u_{\min},\ \forall n,\ m, \tag{22c}$$

$$\|\mathbf{q}_m[n]-\mathbf{q}_k[n]\|\geq d_{\min^2},\ \forall n,m\neq k\ , \tag{22d}$$

$$\mathbf{\Phi}=\left(x_m[n],y_m[n],s_{mj}[n],\lambda_{mj}[n]\right)\geq 0,\forall n,m,j, \tag{22e}$$

$$\text{(12c)–(12h), (14b), (17a), (21b)} \tag{22f}$$

However, due to the nonconvex constraints (22b), (22c), (22d), and (22e), (P7) remains a nonstandard convex optimization problem. To simplify this formulation, we employed the following redundant variable method and the SCA method to iteratively find an approximate solution.

By introducing the redundant variable $\boldsymbol{\tau}=\{\tau_m[n],\forall n,m\}$, the nonconvex constraints (22b) and (22c) can be equivalently expressed as:

$$\delta\sum_{n=1}^{N}\left[c_1||\mathbf{u}_m[n]||^3+\frac{c_2}{\tau_m[n]}\left(1+\frac{||\mathbf{a}_m[n]||^2}{g^2}\right)\right]+\Delta_k\leq\Gamma,\forall m, \tag{23a}$$

$$\|\mathbf{u}_m[n]\|^2 \geq u_{\min}^2,\ \forall n,\ m, \tag{23b}$$

$$\tau_m^2[n] \leq \|\mathbf{u}_m[n]\|,\ \forall n, m, \tag{23c}$$

$$\tau_m[n] \geq u_{\min},\ \forall n, m. \tag{23d}$$

Based on the above transformations, it is evident that constraint (23a) is jointly convex with the variables $\{\mathbf{u}_m[n], \mathbf{a}_m[n], \tau_m[n]\}$, and constraint (23d) is convex with the variable $\tau_m[n]$. However, constraints (23b) and (23c) are nonconvex with the variable $\mathbf{u}_m[n]$. By applying the properties of the first-order Taylor expansion, we can expand $\|\mathbf{u}_m[n]\|^2$ at any feasible point $\mathbf{u}_{m,\text{fea}} \triangleq [\mathbf{u}_{m,\text{fea}}[1], \ldots, \mathbf{u}_{m,\text{fea}}[N]]$, resulting in the following lower bound:

$$\tilde{\mathbf{u}}_m[n] = \|\mathbf{u}_{m,\text{fea}}[n]^2\| + 2(\mathbf{u}_{m,\text{fea}}[n])^{\mathrm{T}}(\mathbf{u}_m[n] - \mathbf{u}_{m,\text{fea}}[n]). \tag{24}$$

Consequently, constraints (23b) and (23c) can be approximated and converted into the following convex constraints:

$$\tilde{\mathbf{u}}_m[n] \geq u_{\min}^2,\ \forall n, m, \tag{25a}$$

$$\boldsymbol{\tau}_m^2[n] \leq \tilde{\mathbf{u}}_m[n], \forall n, m. \tag{25b}$$

For the nonconvex collision constraint (22d), we applied a first-order Taylor expansion to the left-hand side at any feasible point $\mathbf{q}_{m,\text{fea}} \triangleq \left[\mathbf{q}_{m,\text{fea}}[1], \ldots, \mathbf{q}_{m,\text{fea}}[N]\right]$, leading to the following expression:

$$\begin{gathered}\|\mathbf{q}_m[n] - \mathbf{q}_k[n]\|^2 \geq 2\left(\mathbf{q}_{m,\text{fea}}[n] - \mathbf{q}_{k,\text{fea}}[n]\right)^T(\mathbf{q}_m[n] - \mathbf{q}_{\mathbf{k}}[n]) \\ \|\mathbf{q}_{m,\text{fea}}[n] - \mathbf{q}_{k,\text{fea}}[n]\|^2, \forall n, m \neq k.\end{gathered} \tag{26}$$

Therefore, constraint (22d) can be approximately reformulated as the following convex constraint:

$$2\left(\mathbf{q}_{m,\text{fea}}[n] - \mathbf{q}_{k,\text{fea}}[n]\right)^T(\mathbf{q}_m[n] - \mathbf{q}_{\mathbf{k}}[n]) - \|\mathbf{q}_{m,\text{fea}}[n] - \mathbf{q}_{k,\text{fea}}[n]\|^2 \geq d_{\min}^2, \forall n, m \neq k. \tag{27}$$

Finally, for the nonconvex constraint (22e), to eliminate its quadratic terms, we performed a first-order Taylor expansion of the $c_{mj}[n]$ term at any feasible point $\mathbf{x}_{m,\text{fea}} \triangleq [\mathbf{x}_{m,\text{fea}}[1], \ldots, \mathbf{x}_{m,\text{fea}}[N]]$ and $\mathbf{y}_{m,\text{fea}} \triangleq \left[\mathbf{y}_{m,\text{fea}}[1], \ldots, \mathbf{y}_{m,\text{fea}}[N]\right]$. Therefore, the lower bound of the $\widetilde{c}_{mj}[n]$ can be expressed as:

$$\begin{aligned}\widetilde{c}_{mj}[n] = \quad & -x_{m,\text{ teat}}^2[n] + 2x_{m,\text{ fea}}[n]x_m[n] - 2\widetilde{x}_j x[n] + \widetilde{x}_j^2 - y_{m,\text{ fas}}^2[n] \\ & +2y_{m,\text{ fea}}[n]y_m[n] - 2\widetilde{y}_j y[n] + \widetilde{y}_j^2 y[n] + \widetilde{y}_j^2 + H^2 - s_{mj}[n]\end{aligned} \tag{28}$$

Thus, constraint (22e) can be approximately transformed into the following convex constraint:

$$\tilde{\boldsymbol{\Phi}} = \left(x_m[n], y_m[n], s_{mj}[n], \lambda_{mj}[n]\right) \geq 0, \forall n, m, j, \tag{29}$$

where:

$$\tilde{\boldsymbol{\Phi}}(x_m[n], y_m[n], s_{mj}[n], \lambda_{mj}[n]) = \begin{bmatrix} \lambda_{mj}[n]+1 & 0 & \widetilde{x}_j - x_m[n] \\ 0 & \lambda_{mj}[n]+1 & \widetilde{y}_j - y_m[n] \\ \widetilde{x}_j - x_m[n] & \widetilde{y}_j - y_m[n] & -\varepsilon_j^2\lambda_{mj}[n] + \widetilde{c}_{mj}[n] \end{bmatrix}.$$

3.2.4. Summary of the Flight Parameter Optimization

Based on the above transformations, the problem (P4) can be approximately reformulated as follows:

$$\begin{gathered}(\text{P8}) \max_{\{\mathbf{q},\mathbf{u},\mathbf{a},\mathbf{I},\mathbf{L},\mathbf{S},\boldsymbol{\lambda},\boldsymbol{\tau}\}} \frac{1}{N}\frac{1}{M}\sum_{n=1}^{N}\sum_{m=1}^{M} p_m[n]\beta_0\widetilde{\gamma}_{m,\text{fea}}[n] \\ s.t. (12\text{c}) - (12\text{h}), (14\text{b}), (17\text{a}), (21\text{b}), (22\text{d}), (23\text{d}), (25\text{a}), (25\text{b}), (27), (29).\end{gathered} \tag{30}$$

It is evident that the (P8) is a standard convex programming problem that can be efficiently solved using convex optimization tools.

Overall, the above effort converted the initial UAV swarm flight parameter optimization problem (P4) into a standard convex problem (P8) using redundant variables and the SCA method. Problem (P8) served as a lower bound for problem (P4) and is equivalent to it at specific points $(\mathbf{x}_{\text{fea}}, \mathbf{y}_{\text{fea}}, \mathbf{u}_{\text{fea}}, \mathbf{I}_{\text{fea}}, \mathbf{L}_{\text{fea}})$. As the objective function is bounded and nondecreasing during iterations, the algorithm has assured convergence. The process of the algorithm is summarized in Algorithm 1 as follows:

Algorithm 1. Flight Parameter Optimization Algorithm for UAV Swarms

1. **Initialize** $(\mathbf{x}^{(0)}, \mathbf{y}^{(0)}, \mathbf{u}^{(0)}, \mathbf{I}^{(0)}, \mathbf{L}^{(0)})$. Set the maximum number of iterations $T_{1,\max}$ and the threshold ℓ_1. Let $t_1 = 0$.
2. **Repeat**
3. Set $t_1 \leftarrow t_1 + 1$;
4. Set the feasible point $\mathbf{x}_{\text{fea}} = \mathbf{x}^{(t_1-1)}, \mathbf{y}_{\text{fea}} = \mathbf{y}^{(t_1-1)}, \mathbf{u}_{\text{fea}} = \mathbf{u}^{(t_1-1)}, \mathbf{I}_{\text{fea}} = \mathbf{I}^{(t_1-1)}, \mathbf{L}_{\text{fea}} = \mathbf{L}^{(t_1-1)}$, then solve the problem (P8) and refer to the optimal solution as $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{u}^*, \mathbf{I}^*, \mathbf{L}^*)$;
5. Update the variables $\mathbf{x}^{(t_1)} = \mathbf{x}^*, \mathbf{y}^{(t_1)} = \mathbf{y}^*, \mathbf{u}^{(t_1)} = \mathbf{u}^*, \mathbf{I}^{(t_1)} = \mathbf{I}^*, \mathbf{L}^{(t_1)} = \mathbf{L}^*$;
6. **Until** the maximum number of iterations $T_{1,\max}$ is reached or the increment of the objective function is less than the given threshold ℓ_1.

3.3. Overall Description

Based on the above analysis, the summary of the cooperative trajectory planning and power control algorithm for UAV swarms is shown in Algorithm 2. The suboptimal solution to problem (P1) was achieved through alternating the optimization of the power control and the flight parameters, with the solution to each iteration serving as an input for the next. Since the initial points $\left(\mathbf{Q}^{(0)}, \mathbf{U}^{(0)}, \mathbf{A}^{(0)}, \mathbf{P}^{(0)}\right)$ satisfied the constraint conditions, each iteration yielded a feasible solution. Moreover, as shown in the previous two subsections, the optimal values obtained for problems (P2) and (P4) did not decrease. Consequently, the alternating iterative optimization of these two subproblems progressively improved the objective function value of problem (P1) until it stabilized at a feasible solution.

Algorithm 2. Cooperative Trajectory Planning and Power Control Algorithm

1. Construct the initial feasible solution $\left(\mathbf{Q}^{(0)}, \mathbf{U}^{(0)}, \mathbf{A}^{(0)}, \mathbf{P}^{(0)}\right)$. Set the maximum number of iterations $T_{2,\max}$ and the threshold ℓ_2. Let $t_2 = 0$.
2. **Repeat**
3. Given $\left(\mathbf{Q}^{(t_2)}, \mathbf{U}^{(t_2)}, \mathbf{A}^{(t_2)}\right)$, determine the optimal solution $\mathbf{P}^*$ to problem (P2). Let $\mathbf{P}^{(t_2+1)} = \mathbf{P}^*$;
4. Given $\mathbf{P}^{(t_2+1)}$, determine the optimal solution $(\mathbf{Q}^*, \mathbf{U}^*, \mathbf{A}^*)$ to problem (P4). Then, let $\mathbf{Q}^{(t_2+1)} = \mathbf{Q}^*$, $\mathbf{U}^{(t_2+1)} = \mathbf{U}^*$, and $\mathbf{A}^{(t_2+1)} = \mathbf{A}^*$;
5. Update $t_2 \leftarrow t_2 + 1$;
6. **Until** the maximum number of iterations $T_{2,\max}$ is reached or the increment of the objective function is less than the given threshold ℓ_2.

4. Numerical Results

In this section, we simulate a multi-UAV task execution scenario where the UAV swarm flies from a certain starting point to the endpoint at a fixed altitude, maintaining communication with the GCU throughout the whole flight. In addition, we assume the mission duration $T = 30$ s is divided into equal time slots $\delta = 1$ s. The GCU is positioned at specific coordinates $(0, 0, 0)$ m with a specified transmit power $p_0 = 10$ W. Meanwhile, $J = 3$ jammers are positioned on the ground, with estimated horizontal coordinates at $\tilde{\mathbf{w}}_{j,1} = (700, 800)$ m, $\tilde{\mathbf{w}}_{j,2} = (500, 600)$ m, and $\tilde{\mathbf{w}}_{j,3} = (300, 200)$ m. The estimated transmit power for each jammer is the same value $\widetilde{p}_{j,1} = \widetilde{p}_{j,2} = \widetilde{p}_{j,3} = 0.1$ W, and their power estimation errors are uniformly set to $\xi_m = 0.02$ W. For simplicity, unless specified otherwise, all relevant simulation parameters are listed in Table 1.

Table 1. Simulation parameters.

Description	Parameter and Value
UAV maximum speed	$V_{\max} = 50$ m/s
UAV minimum speed	$V_{\max} = 3$ m/s
UAV maximum acceleration	$a_{\max} = 5$ m/s^2
Channel gain	$\beta_0 = -30$ dB
Noise power	$\sigma^2 = -80$ dBm
Path-loss exponent	$\alpha = 2$
Maximum tolerance	$\ell = 10^{-4}$
Energy consumption parameter	$c_1 = 9.26 \times 10^{-4}$, $c_2 = 2250$
Gravitational acceleration	$g = 9.8$ m/s^2

Next, we showcase the benefits of our proposed robust multi-UAV cooperative trajectory planning and power control scheme (referred to as the robust scheme) when navigating environmental uncertainties. Meanwhile, we introduce the following two benchmark algorithms for comparison:

(1) Fixed trajectory scheme: in this scheme, we assume that each UAV flies in a straight line from the initial point to the final point at a constant velocity.

(2) Nonrobust scheme: in this scheme, we assume the estimated information on each jammer to be accurate, i.e., $\Delta\mathbf{w}_{j,m} = 0, \Delta p_{j,m} = 0, \forall m$, and then compute the worst-case average SINR of the multi-UAV system.

In both benchmark algorithms, we boldly set the UAVs' energy to be enough to ensure that each UAV can finish the corresponding mission under each algorithm.

4.1. Convergence Evaluation

First, our experiment verified the convergence of the proposed algorithm compared with the nonrobust scheme The iteration processes of the robust scheme and nonrobust scheme are shown in Figure 2. From the figure, it can be observed that, in the scenario with the jammer location uncertainty radius $\epsilon_m = 10$ m, the objective function values of both algorithms increased rapidly with the number of iterations. However, the nonrobust scheme converged after four iterations, whereas the robust scheme required five iterations to converge. This is because the robust scheme needs additional computational effort to address the semi-infinite constraints arising from the uncertainty in the jammer information, whereas the nonrobust scheme does not have to account for this constraint, making it easier to converge.

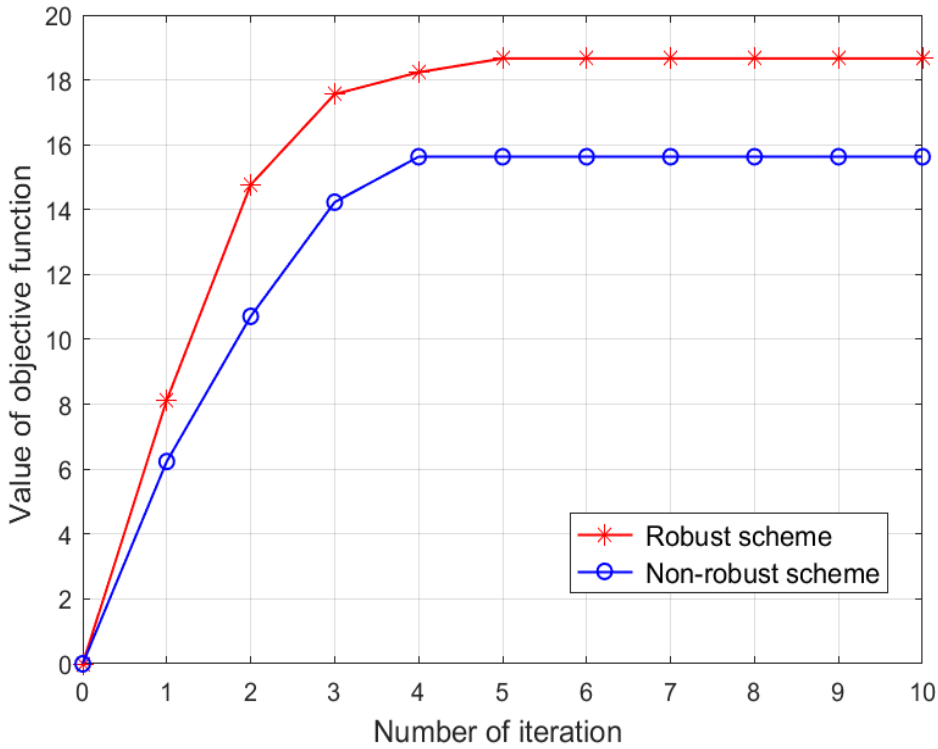

Figure 2. Convergence of the algorithms ($\varepsilon_m = 10$ m).

4.2. Trajectory Result

Next, we modeled the multi-UAV trajectory curves under three different algorithms. Figure 3 depicts the UAV trajectories in a scenario with a jammer location uncertainty radius of $\epsilon_m = 40$ m, considering different energy constraints and algorithms. It can be seen that, compared to the nonrobust scheme, the UAV trajectories under the proposed robust scheme were farther away from the jammers. This is because increasing the distance between the UAVs and the jammers reduces the interference, thereby improving communication quality with the GCU. This demonstrates that the proposed robust scheme provides a more reasonable trajectory planning for the UAV swarm. Additionally, it can be seen that, with the same algorithm, UAV trajectories were farther from the jammers when the energy consumption limit was higher ($\Gamma = 8000$ J) compared to a lower energy limit ($\Gamma = 5000$ J). Clearly, higher energy allowances enabled the UAVs to find better trajectories to counteract the interference, something which is intuitively validated by this experiment.

Figure 4 compares the UAV trajectories under different algorithms and under varying jammer uncertainty radii when the UAV energy limit is set as $\Gamma = 5000$ J. The results show that the flight paths of the UAVs using the proposed robust scheme were superior to those of the two benchmark algorithms. Additionally, the analysis indicates that, regardless of the value of the jammer's uncertainty radius ϵ_j, the UAV trajectories under the nonrobust scheme remained constant. In contrast, as ϵ_j increased, the distance between the UAVs and the jammers increased, with the UAVs moving further away from the jammers under the robust scheme. This experiment demonstrates the effectiveness of the proposed robust scheme in planning UAV swarm trajectories under conditions of jammer uncertainty.

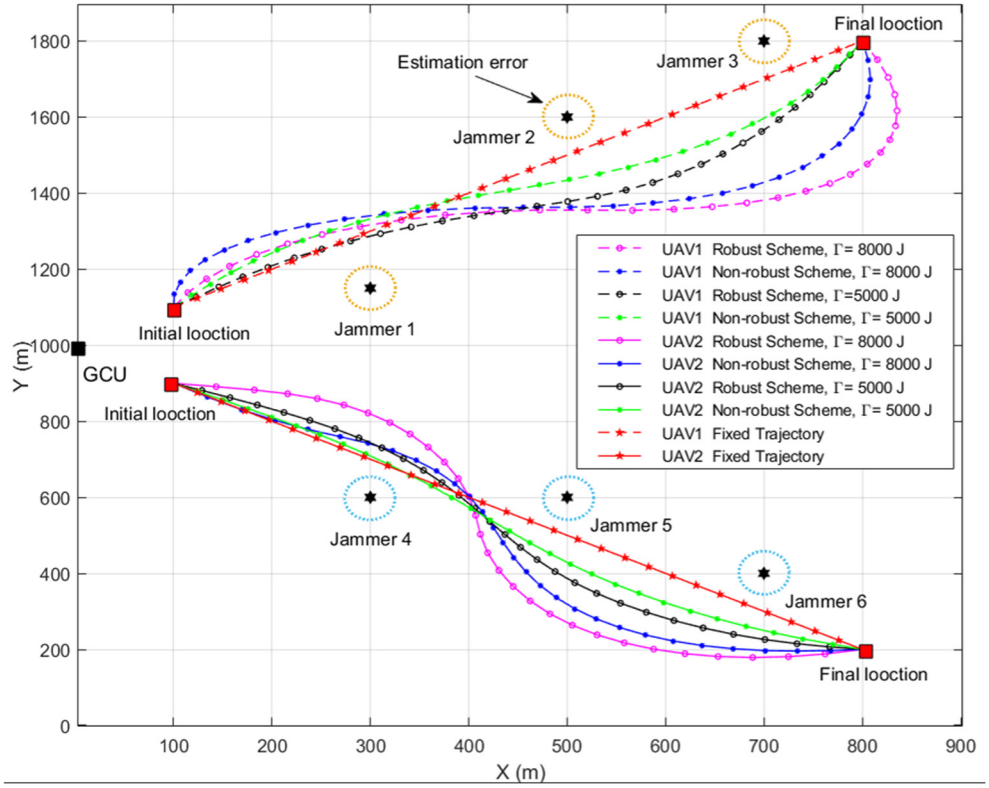

Figure 3. The trajectory of a UAV for $\varepsilon_m = 40$ m for various values of Γ.

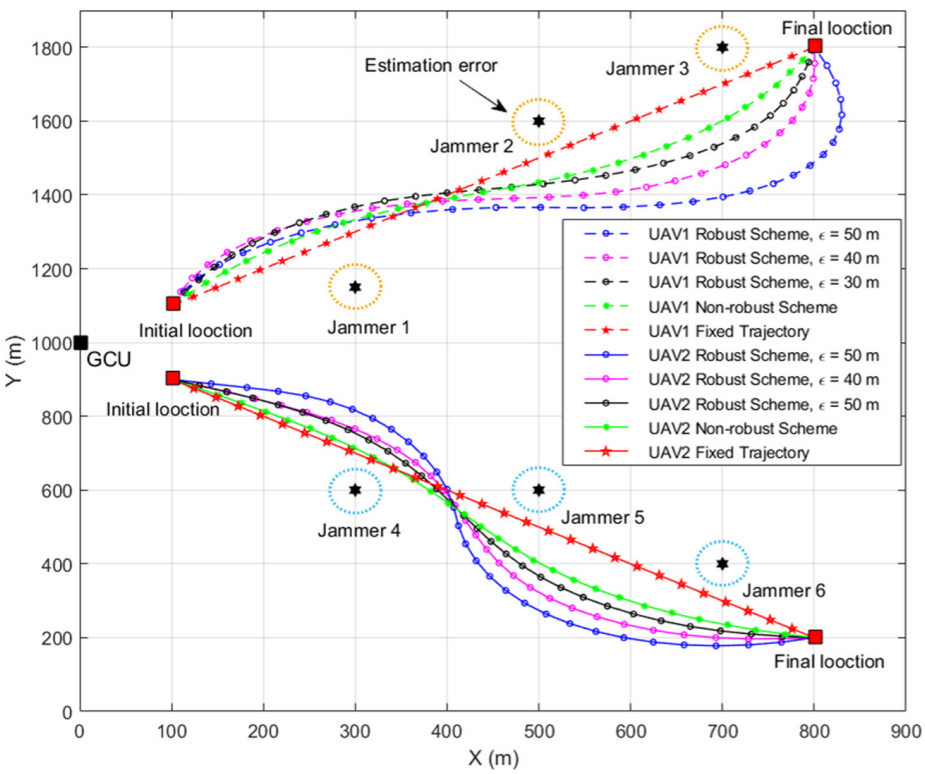

Figure 4. The trajectory of a UAV for Γ = 5000 J under varying uncertain jammer radii.

Figure 5 shows the uppermost flying UAV from Figure 3 and depicts its velocity variations. From the figure, it is evident that, for both the robust scheme and the nonrobust scheme, UAVs initially started with relatively low speeds to maintain high-quality communication with the GCU at close distances. As the UAVs progressively distanced themselves from the GCU and got closer to the jammers, their communication quality was significantly affected. At this point, the UAVs altered their flight path to circumvent interference (as shown in Figure 3) and increased their speed to move farther away from the jammers. This demonstrates that our simulation aligns well with practical scenarios.

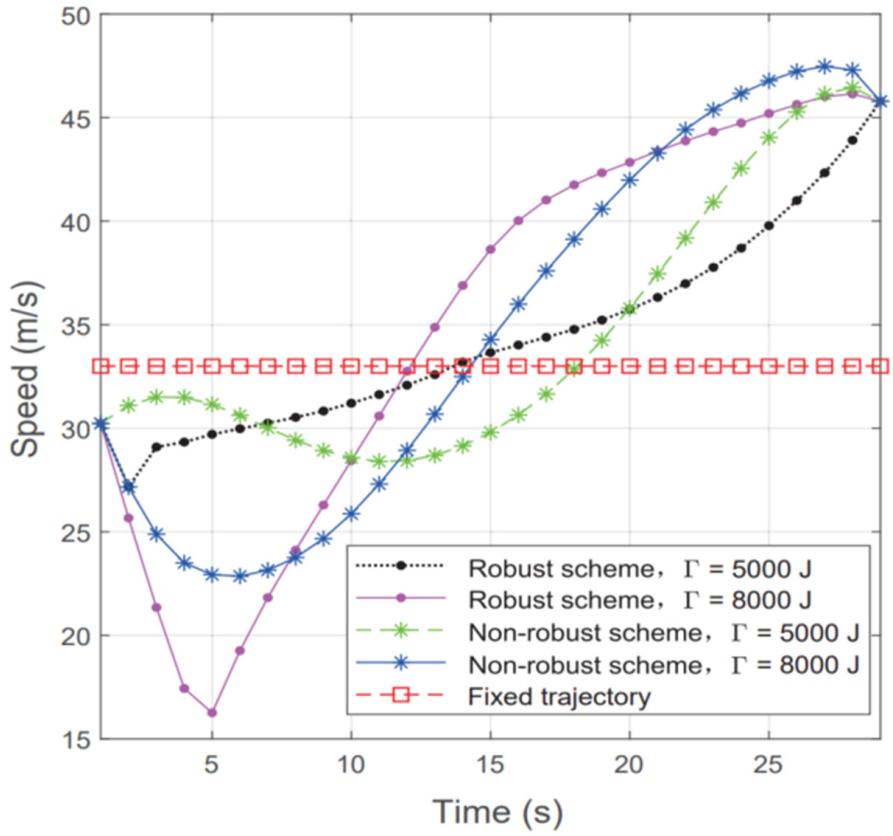

Figure 5. The speed of a UAV for $\epsilon_m = 40$ m under different values of Γ.

4.3. System Performance

In what follows, Figure 6 compares the performances of the robust scheme and the nonrobust scheme in terms of average worst-case SINR for the UAV swarm at various energy consumption levels. It can be observed that, compared to the nonrobust scheme, the proposed robust scheme enabled the UAV swarm to achieve a higher average worst-case SINR, approximately 115% higher. The analysis indicates that this improvement in swarm performance can be attributed to the robust scheme optimizing UAV flight trajectories more effectively, thereby reducing external interference and enhancing communication quality. Furthermore, it is observed that, as the energy consumption limit Γ increased, the communication performance of the swarm gradually improved. However, beyond a certain threshold, increasing Γ no longer enhanced the communication performance of the UAV swarm under different algorithms. This occurred because, when the UAVs have a lower energy consumption limit, increasing Γ allows them to use more energy to explore better trajectories, thereby improving SINR. Yet when Γ becomes sufficiently large, the optimal trajectory for the UAV swarm requires less energy than the current limit, meaning that further increasing Γ does not enhance communication performance.

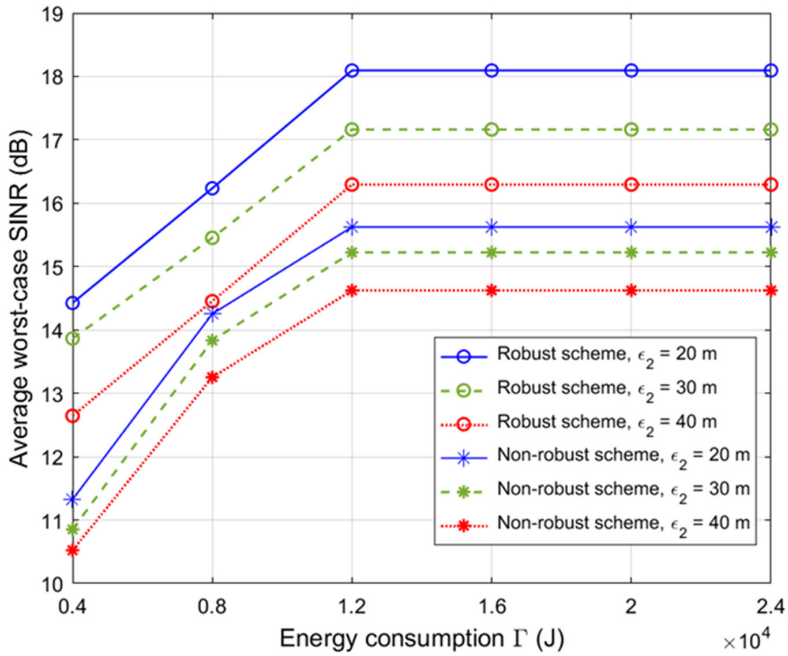

Figure 6. Average worst-case SINR relative to the energy consumption threshold Γ.

Finally, we compared the communication performance of the UAV swarm under different jammer uncertainty radii, as depicted in Figure 7. We noticed that, under both the robust scheme and the nonrobust scheme, the communication performance of the swarm decreased as the uncertainty radius of the jammers increased. As shown in Figure 4, even though the trajectory of the UAV swarm remained unchanged under the nonrobust scheme across varying uncertainty radii, the computed average worst-case SINR still decreased with an increasing uncertainty radius of the jammers. Moreover, due to its neglect of uncertainty in jammer locations, the nonrobust scheme failed to account for the impact of uncertainty on swarm trajectory planning, whereas the robust scheme significantly enhanced the performance by addressing these uncertainties comprehensively.

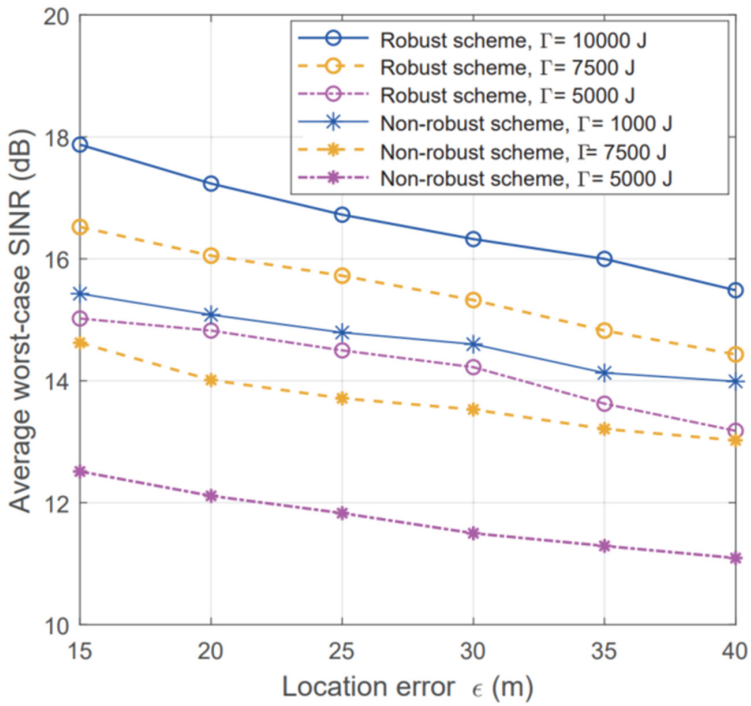

Figure 7. Average worst-case SINR relative to the estimated jammer locations.

4.4. Overall Analysis

Overall, we validated the effectiveness of the proposed algorithm in three key aspects:

1. Convergence analysis: we conducted simulations to evaluate the convergence curves for both robust and nonrobust cases, demonstrating the convergence properties of our algorithm.
2. Real-world scenario simulation: we simulated actual flight scenarios with uncertain jammer locations and varying energy consumptions, confirming the algorithm's ability to effectively plan UAV trajectories under different levels of uncertainty in jammer information.
3. Performance evaluation: we generated performance graphs across various parameters, verifying the superiority of our algorithm with respect to the enhancement of the communication performance in uncertain environments when compared to the benchmark algorithms.

In conclusion, the proposed algorithm provides a theoretical basis for global planning and autonomous decision-making of UAV swarms under uncertain adversary information, thereby offering robust support for next-generation electromagnetic spectrum warfare systems.

5. Conclusions

This paper addresses the design of trajectory planning and power allocation for efficient control and reliable communication in GCU-controlled UAV networks. In this scenario, the GCU manages multiple UAVs, making them complete tasks in the presence of numerous jammers, despite having incomplete information about their locations and power levels. To improve the communication performance in multi-UAV networks, we formulate a joint optimization problem that involves power allocation and flight parameter planning for UAVs, aiming to maximize the average worst-case received SINR. Furthermore, we propose an effective iterative algorithm to tackle this problem and conduct simulation experiments focusing on three key aspects: algorithm convergence analysis, real-world scenario simulation, and performance evaluation across various parameters. The results clearly demonstrate the effectiveness of our algorithm.

For future research, we plan to extend the current model to accommodate more complex environmental conditions and mission scenarios. Additionally, integrating machine learning techniques could improve the adaptability of the optimization algorithms, enabling real-time adjustments in dynamic environments. Another promising direction is the application of integrated sensing and communication (ISAC) to enhance the UAVs' ability to process and respond to complex environmental stimuli in real-time.

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